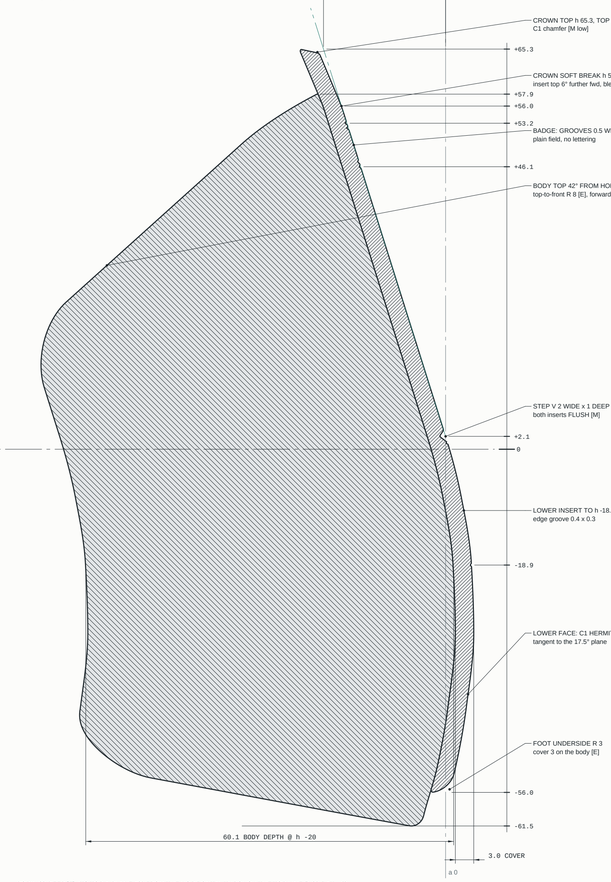
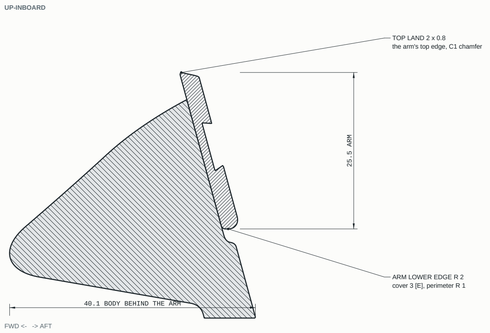


BODY: 60 DEEP [E], ITS TOP A PLANE 42° FROM THE HORIZONTAL THROUGH THE TOP LINE (h 60.5 AT u 0)
forward edges R 8, top-to-front R 8, foot R 14 [E]; black body and grips one moulding
PLANE u 0 (CUTTING PLANE Z-Z ON L6Y FRONT ELEVATION)

SECTION Z-Z

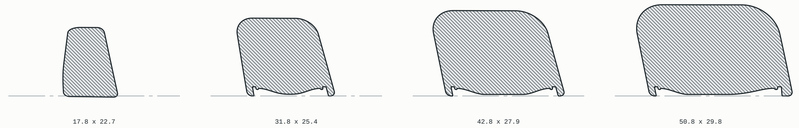
YOKE CENTRELINE u 0, FROM OUTBOARD, NOSE LEFT - 2:1



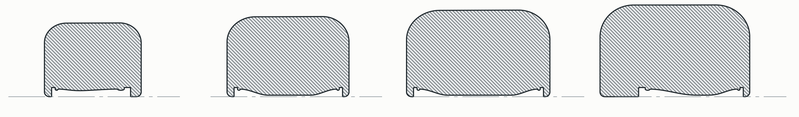
SECTION K-K

THROUGH (52, 45) NORMAL TO THE ARM (30° UP OUTBOARD), LOOKING INBOARD (L6Y FRONT ELEVATION) - 2:1

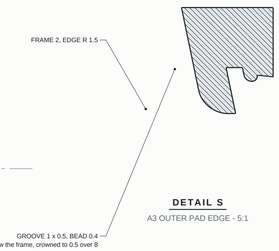
OUTBOARD GRIP, NORMAL TO THE AXIS, LOOKING DOWN IT (FWD UP, PAD DOWN; OUTBOARD LEFT)



INBOARD GRIP, NORMAL TO THE AXIS, LOOKING DOWN IT (FWD UP, PAD DOWN; INBOARD RIGHT)

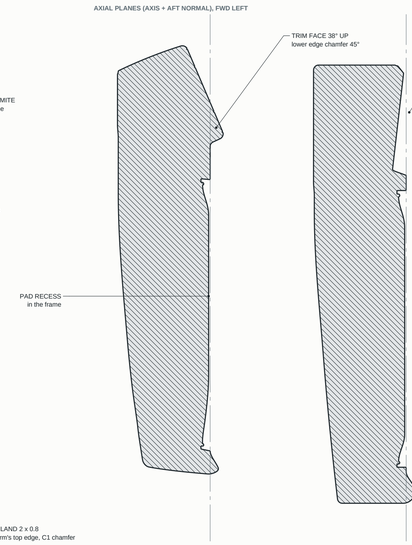


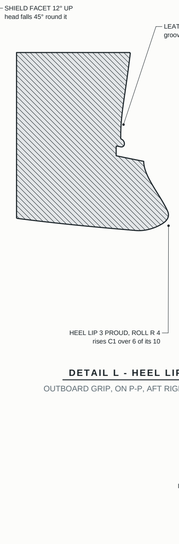
PAD: leather 2.5 below the 2 frame, crowned to 0.5 below over 8; groove 1 x 0.5 + bead 0.4; frame roll R 1.5; forward R 11 [E] <= 0.45 x half-width; depth PCHIP (grip_depth) [M low]



DETAIL S

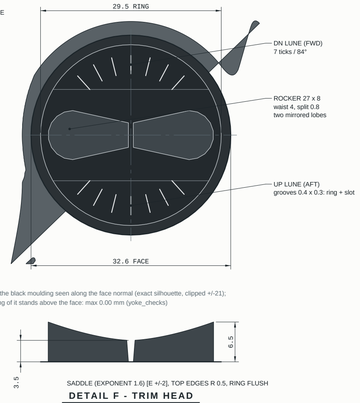
A3 OUTER PAD EDGE - 5:1





DETAIL L - HEEL LIP

OUTBOARD GRIP, ON P-P, AFT RIGHT - 3:1

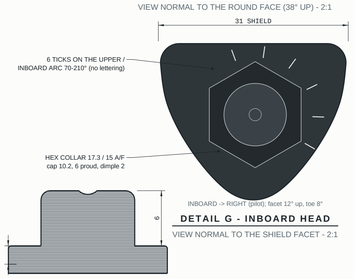


grey: the black moulding seen along the face normal (exact silhouette, clipped +/-21);
nothing of it stands above the face: max 0.00 mm (yoke_checks)

SADDLE (EXPONENT 1.6) [E +/-2], TOP EDGES R 0.5, RING FLUSH

DETAIL F - TRIM HEAD

VIEW NORMAL TO THE ROUND FACE (38° UP) - 2:1



INBOARD -> RIGHT (pitot), facet 12° up, toe 8°

DETAIL G - INBOARD HEAD

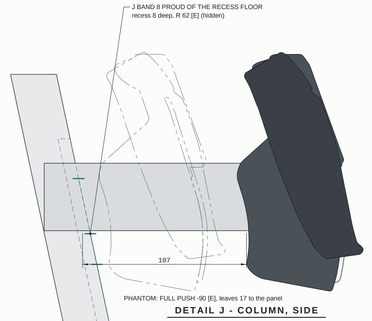
VIEW NORMAL TO THE SHIELD FACET - 2:1



PLAIN FIELD (NO LETTERING); GROOVES 0.5 x 0.3 CUT IN THE INSERT

DETAIL H - BADGE

ABSTRACT WINGED PLATE, TOP AT h 53.2 - 2:1

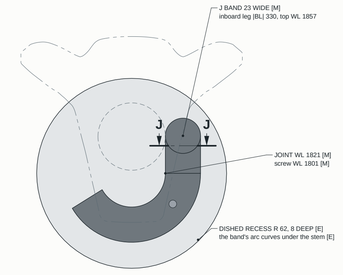


DETAIL J - COLUMN, SIDE

AT THE LOWER PANEL, FROM OUTBOARD - 1:2

SECTION J-J - BAND

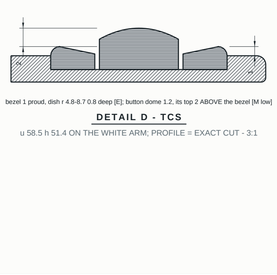
ACROSS THE VERTICAL LEG (P1046410 / 09) - 2:1



phantom: the shell; hidden: the column dia 44

DETAIL J - SURROUND, FRONT

LOOKING FORWARD (PILOT; INBOARD RIGHT) - 1:2



bezel 1 proud, dish r 4.8-8.7 0.8 deep [E]; button dome 1.2, its top 2 ABOVE the bezel [M low]

DETAIL D - TCS

u 58.5 h 51.4 ON THE WHITE ARM; PROFILE = EXACT CUT - 3:1

boss 1.5 proud of the body [E], 8 deep: its back free over the hand slot; edge R 0.8, cap top R 1

DETAIL E - AFCS DISCONNECT

u 60.6 h 23.4; BOSS = A SOLID OF THE MOULDING; PROFILE = EXACT CUT - 3:1

PHOTO CHECK (rev K): THE DRAWN SOLIDS RAY-CAST THROUGH THE PHOTO CAMERAS (mm, + = drawn wider)

PHOTO (CAMERA)	COVER LM	HEADS	GRIP OUT: OUTER / SLOT (mean, max)	GRIP IN: OUTER / SLOT (mean, max)	PADS mean (max)
P1046408 pilot (panel_408)	1.4 / 2.7	2.1 / 1.4	+0.3, +4.6 / +0.1, +3.0	-0.1, -6.5 / -1.2, -3.9	out 0.7 (2.6); in 0.8 (1.7)
P1046408 co-pilot (panel_408)	1.8 / 3.9	2.9 / 0.8	+1.4, +6.6 / -0.2, -2.1	-0.9, -3.5 / +0.7, +4.1	out 1.7 (3.4); in 0.8 (2.2)
P1046410 pilot	0.8 / 1.6	5.4 / 0.9	-0.3, +7.3 / -0.6, -5.4	-1.2, -8.4 / +3.3, +7.4	-
Mexico press, pilot	1.7 / 3.9	1.7 / 1.9	+2.7, +5.5 / +1.1, +1.5	-2.6, -8.6 / +1.6, +2.4	in 2.3 (5.9)
Mexico press, co-pilot	1.9 / 3.1	0.9 / 0.9	+0.3, +6.7 / -0.1, +0.9	+1.3, +4.0 / -7.4, -20.8	out 1.1 (2.7)
AOPA press, pilot (15 deg from outboard)	1.9 / 4.0	1.9 / 0.2	- / -	- / -	out 5.6 (15.9); in 2.9 (9.1)
AOPA press, co-pilot (~50 deg from inboard)	1.6 / 3.7	4.4 / 13.1	-11.2, -22.1 / -	- / -8.8, -14.9	in 9.4 (19.6)
rev J: P1046408 L (judges r2)	-	-	slot +4.0 (h 14) , -2.8	-	out 0.7; in 0.5

GRIP SIDES PER BAND OF h: SIGNED MAX mm (h -50..-30 | -30..-10 | -10..10 | 10..30)

SIDE	P408 L	P408 R	P410 L	MEX L	MEX R	AOPA L	AOPA R
OUT SLOT	+3.0 -1.2 +1.3 +1.5	-2.1 -1.3 +1.2 +1.3	+2.9 -1.2 -0.7 -5.4	- , +1.5 +1.2	+0.9 -0.7 -0.7 -0.8	-	-
OUT OUTER	+2.8 -0.8 -0.3 -0.8	+3.6 +1.1 , -	+2.7 +3.1 -1.5 -0.6	- , -	-2.1 , -5.5 -0.7	-	-22.1 -17.5 -13.4 -12.2
IN SLOT	-2.1 -3.9 -2.7 +1.0	+0.8 -1.6 +2.5 +4.1	+2.5 +3.9 +6.2 +7.4	+1.4 +2.4 +2.2	- -6.6 -2.4 -11.5	-	- -12.5 -11.5 -8.6
IN OUTER	-6.5 -5.9 -1.2 -0.6	-3.5 -2.5 -1.5 -1.3	-6.1 -8.4 -6.5 +0.5	+5.6 -8.6 -7.3 -8.0	+4.0 +2.0 +1.7 +2.2	-	-

GRIP PLANES (judges FID2-01): rake / toe / a0 / wall direction fitted per grip to these curated rows and the pad traces of all 7 views (the grip prism, defined in its plane coordinates, re-mapped exactly for each plane). The 5 views from inboard (P1046408 L / R, P1046410, Mexico L / R: 23-27° off aft, Mexico 15-20° from above) give outboard 19.3° / 19.4° / -3.7 / 0.61, inboard 20.5° / 18.4° / -4.9 / 0.03 -- the table's 20 / 18 / -3.75 / 0.6, 0 kept; their cost rises 10 % at rake +/-3° (out) / +/-5° (in), a0 +/-3 / +/-5. With the AOPA views the fit moves < 1.5° / 2.5 and the 7-view cost drops only 7 / 3 %: no plane the 5 views allow meets the AOPA residuals (co-pilot: outboard outer edge -9, -16 below h 0, inboard slot -10, -12, inboard pad 9; pilot: pads 5.6 / 2.9) -- the same view puts the inboard button 13 off (6 views within 1-2) and its pose is degenerate along the line of sight: reported, not fitted (L6Y open point 2).

INBOARD NECK (criterion, P1046408): the drawn moulding's top edge vs the photo per pixel column, arm tip to head -- pilot mean -0.6, jmean| 2.3, max -6.9 mm (against the red placard; rev I 'neck in' 14.2 / 20.7); co-pilot -4.5, max -7.5 (drawn LOW near the head; against the dark PFD, low contrast).

Pose per photo fitted to the COVER landmarks only (TCS, AFCS, badge, step ends, lower-insert corners, foot; the AOPA co-pilot view with its step / insert picks); heads, grips, pads, neck and shell are checks. Grip rows: the CURATED photo silhouettes -- auto edges per image row in the near views with the panel recesses masked, the inboard slot sides hand-traced and snapped past the J band / panel seam, the AOPA co-pilot sides hand-traced; the AOPA pilot auto rows dropped (they lock onto the leather on the dark panel) -- against the ray-cast grip's own hits (part-labelled), per image row, h on the grip axis. Overlays: refs/cache/overlays/yoke (private).

CORNERS & RADII (every other sample is on a C1 spline)

OUTLINE	N	CORNERS: SAMPLE #, (u, h), RADIUS (curve_law)
shell (white cover), half	49	#4 (22, 62.8) R 3; #10 (33.7, 52) R 1.5; #21 (66, 60.2) R 2; #22 (72.3, 54.8) R 2; #23 (72.3, 47.8) R 2; #24 (69, 44.6) R 2; #45 (20, -55.3) R 6
lower insert, half	8	#0 (0, 1) R 0; #1 (12.6, 3) R 0.5
outboard grip aft face	37	#6 (130.5, 93.6) R 4; #20 (136.6, -30.4) R 5; #23 (121.2, -38) R 4
inboard grip aft face	37	#2 (91.4, 84) R 4; #8 (118.5, 87.6) R 6; #10 (125.6, 60) R 3; #21 (140.5, -36) R 5; #24 (118.5, -45.8) R 4
outboard pad	29	#0 (125.5, 60) R 3; #7 (91, 38) R 5; #16 (123.5, -28.4) R 3; #19 (134.8, -25.5) R 4
inboard pad	31	#0 (127.8, 55.3) R 3; #7 (89.4, 32) R 4; #17 (116, -34.4) R 2; #21 (139, -29.6) R 4
shield facet	9	#0 (-15.5, 12) R 3; #1 (15.5, 12) R 3
insert corners	-	junction R 0; step_end R 0.5
cover edges	-	perimeter R 1; arm lower R 2; foot underside R 3; tip facet corners R 2
grip edges	-	frame roll R 1.5; forward R 11 [E] capped at 0.45 x the inscribed half-width; heel R 4; trim face R 1.5; pad step crisp, groove 1 x 0.5, bead 0.4
body edges	-	aft R 2; forward R 8 [E]; top-to-front R 8 [E]; foot R 14 [E]; neck-to-head blend R 3; AFCS boss R 0.8, blend R 1.5
cover features	-	crown soft break R 20; top land C1 chamfer 2 x 0.8; TCS seat 1.5 deep

NOTES

1. CURVE LAW: centripetal Catmull-Rom (alpha 0.5) through the table samples, 0.25 sampling; corners as listed (R 0 sharp, R > 0 tangent fillet); YOKE_DESIGN curve_law; the solids use the EXACT distance to these polylines.
2. ACCEPTANCE (model/yoke_checks.py, judges DR2-01..04): fields: zero level vs the curve-law polylines, max shell 0.0358, grip_out 0.0320, grip_in 0.0342, pad_out 0.0312, pad_in 0.0223, body 0.0257 mm; nothing above the trim face: max 0.00 mm; above the shield facet: max 0.00 mm; untabled cusps (> 25 deg within +-0.15 mm): grip_out 0 / 41 sections, grip_in 0 / 39 sections, Z-Z 0, K-K 0; vertex normals vs analytic (0.25 mm depth-map mesh, 95th / 99th / max deg): cover stem / inserts 0.00 / 0.06 / 21.2; cover arm + TCS 0.29 / 1.13 / 7.3; outboard grip pad / frame 0.51 / 1.53 / 29.3; inboard grip pad / frame 0.49 / 1.75 / 4.3.
3. SECTIONS: exact contours of the solids (model/yoke_geom.py) on the cutting planes; the black body and the grips are one moulding (smooth union R 3), the white cover a separate part; hatching per part.
4. Sources per value: the [M] / [E] / (S] tags of YOKE_DESIGN (printed in model/interior.py).

REVISIONS - PC12-L6Y2

REV	DATE	DESCRIPTION
A	2026-10-05	first issue with L6Y rev B (judges r1 D1 / D2 / D5 / D10, FID1-04..06 / 09..12): exact sections of the rev J solids -- centreline, arm, 8 grip sections, the grips' axial profiles with the heel lip; the round trim face and the shield facet; switch profiles; the column surround in front view; corner / radius table; photo check per grip row and the inboard neck
B	2026-10-05	rev K (judges r2): exact-field solids; Z-Z / K-K dimensioned with ordinates; grip sections free of untabled cusps (forward R capped; PCHIP depth); pad groove + bead and heel lip R 4 built (details S, P); detail F from the solid (the web held under the face); D / E / G profiles exact cuts of the TCS (button proud), AFCS boss + cap and button solids; section J-J of the column band; acceptance checks; photo check on curated silhouettes incl. the AOPA co-pilot view, part-labelled ray-cast, per-band maxima; grip planes joint-fitted to 7 views (the table confirmed by 5, the AOPA photo open)

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - SECTIONS & DETAILS

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y2	B	AS SHOWN	A1 · 2 OF 2	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-10-05

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION